

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

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Hash Table:

Write a program to:

- a. Implement a hash table with separate chaining for collision handling.

code

size = 5

table = [[] for _ in range(size)]

def insert(key,value):

 table[(key % size)].append((key,value))

insert(10,"roshan")

insert(15,"nitesh")

insert(20,"sunny")

insert(29,"kaushal")

insert(33,"aniket")

print(table)

output

```
[[ (10, 'roshan'), (15, 'nitesh'), (20, 'sunny') ], [], [], [(33, 'aniket')], [(29, 'kaushal')]]
```

b. Store and retrieve data from the hash table.

code

```
size = 5
table = [[] for _ in range(size)]

def insert(key,value):
    table[(key % size)].append((key,value))

def search(key):
    for k ,v in table[key%size]:
        if k==key:
            return v
    return "key not found"

insert(10,"roshan")
insert(15,"nitesh")
insert(20,"sunny")
insert(29,"kaushal")
insert(33,"aniket")

print(table)
key=int(input("enter your key :"))
print(search(key))
```

output

```
----- RESTART: C:/Users/101ad/Desktop/roshan sylv/pract/qs.py -----
[[ (10, 'roshan'), (15, 'nitesh'), (20, 'sunny')], [], [], [(33, 'aniket')], [(29, 'kaushal')]]
enter your key :10
roshan
```